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Ground-shaking-informed regime partitioning for characterizing earthquake-induced secondary hazards

Feng Lin[1,2], Xie Hu[1,2*], Xiao Yu[3], Yueren Xu[4] and Ni An[5]

Abstract

Background Earthquake-induced landslides and liquefaction often occur within the same seismic event but may be controlled by different combinations of ground shaking, topography, hydrology, site conditions, and tectonic setting. This study develops an interpretable remote-sensing framework to map and compare these secondary hazards after the 2025 Dingri Ms 6.8 earthquake in the southern Tibetan Plateau.

Results Multi-source optical and SAR indicators, seismic variables, and geo-environmental factors were integrated using ensemble machine-learning models. A peak ground acceleration (PGA)-guided multivariate Gaussian mixture model was used to stratify hazard samples into PGA-associated statistical regimes, and regime-wise XGBoost models were interpreted using SHAP. The results show that landslide predictions are mainly associated with shaking intensity and topographic conditions, whereas liquefaction predictions show stronger regime-dependent associations with site conditions, hydrological proximity, geomorphic setting, and near-fault effects.

Conclusions The proposed framework provides a practical way to compare model-inferred feature associations across different shaking and environmental backgrounds. The results highlight that coseismic secondary hazards in high-altitude tectonic regions are not governed by a single uniform relationship, but by spatially heterogeneous combinations of seismic and environmental conditions associated with hazard occurrence.

Keywords Landslides, Liquefaction, Earthquake, Remote sensing, Gaussian mixture model, SHAP, Susceptibility mapping

*Correspondence:
Xie Hu
hu.xie@pku.edu.cn
[1]College of Urban and Environmental Sciences, State Key Laboratory of Vegetation Structure, Function and Construction (VegLab), and Key Laboratory for Earth Surface Processes of the Ministry of Education, Peking University, Beijing, China
[2]Beijing Key Laboratory of Spatio-temporal Perception and Urban Resilience, Beijing, China
[3]Illinois State Water Survey, University of Illinois Urbana-Champaign, Champaign 61820, USA
[4]Key Laboratory of Earthquake Prediction, Institute of Earthquake Forecasting, China Earthquake Administration, Beijing 100036, China
[5]College of Civil Engineering and Architecture, Zhejiang University, Zhejiang 310058, China

Introduction

Earthquake-induced secondary hazards, such as landslides and liquefaction, usually pose more severe threats to infrastructure, ecosystems, and human safety than the earthquake itself (Reichenbach et al. 2018). Advances in remote sensing have enabled rapid, large-scale mapping of these hazards using multi-source satellite observations, and machine-learning approaches have become increasingly popular for capturing complex, nonlinear relationships between hazard occurrence and environmental conditions (Dong & Shan 2013; Fan et al. 2019). Despite their success in improving predictive performance, most existing data-driven studies implicitly assume that hazard mechanisms are spatially uniform within a single earthquake event, treating landslides or liquefaction as governed by a common set of controls across the affected region (Xu et al. 2022). However, both geophysical theory and post-earthquake observations suggest that earthquake-induced hazards often arise from multiple coexisting mechanisms, driven by shaking intensity, site conditions, tectonic structures, geomorphological, and hydrological settings. Ignoring this heterogeneity can obscure key process-level differences and limit the physical interpretability of model results derived from remote sensing data. Addressing this challenge requires not only accurate hazard mapping, but also a regime-aware framework capable of resolving and interpreting spatially variable hazard controls within a single earthquake event.

Remote sensing and machine-learning techniques have proven effective for mapping earthquake-induced landslides and liquefaction over large spatial extents (Chen et al. 2026; Shao & Xu 2022; Song et al. 2025a). A wide range of models, including random forest, support vector machines, gradient boosting, and deep neural networks, have been applied to relate hazard occurrence to seismic parameters, topographic attributes, geological settings, and surface indicators derived from optical and Synthetic Aperture Radar (SAR) imagery (Yu et al. 2024). These approaches have substantially improved the spatial completeness and efficiency of post-earthquake hazard inventories and damage assessment, particularly in remote or inaccessible regions (Dong & Shan 2013; Li et al. 2025; Song et al. 2025a; Xu et al. 2022).

Recent studies have introduced interpretable machine-learning techniques, such as feature importance analysis and Shapley Additive exPlanations (SHAP), to improve the transparency of hazard models (Khan et al. 2025; Lundberg & Lee 2017; Zhang et al. 2023; Zhou et al. 2022). While these methods quantify the relative contributions of input variables, they are typically applied to spatially aggregated samples and therefore represent averaged effects across dynamic environments. In the context of earthquake-induced hazards where triggering mechanisms vary substantially with ground-motion intensity, geomorphology, site conditions, and hydrological settings, such global explanations may obscure physically distinct processes. As a result, SHAP-based interpretations derived from mixed samples can conflate multiple coexisting mechanisms, limiting their ability to reveal process-level controls and reducing their geoscientific interpretability. This highlights a critical gap: current interpretable machine-learning approaches lack a mechanism-aware framework that can explicitly resolve spatial heterogeneity in hazard controls (Chen et al. 2018; Kakogeorgiou & Karantzalos 2021; Sun et al. 2022; Tehrany et al. 2019).

The southern Tibetan Plateau provides an ideal natural laboratory to investigate this challenge due to its complex tectonic setting, extreme elevation gradients, and diverse geomorphic and hydrological environments (Hou et al. 2020; Xu et al. 2025a, b; Yang et al. 2025). On January 7, 2025, an Ms 6.8 earthquake struck Dingri County (87.45°E, 28.50°N) at a depth of ~ 10 km, triggering thousands of landslides and widespread liquefaction. By 8:00 a.m. on January 14, 2025, 3,614 aftershocks were recorded, with the largest reaching Ms 5.0 according to the United States Geological Survey (USGS). The event caused 126 fatalities and hundreds of injuries, and exposed ~ 60,000 people to earthquake-related hazards (Fig. 1). The region is located within the Himalayan–Tibetan orogenic system, where ongoing convergence between the Indian and Eurasian plates drives strong crustal deformation and active faulting (Armijo et al. 1986; Chen et al. 2024; Gan et al. 2007; Liang et al. 2013). Such tectonically complex conditions, combined with strong spatial variability in topography and surface environments, facilitate the coexistence of multiple hazard mechanisms within a single earthquake event, making it particularly suitable for mechanism-resolved analysis.

To address these challenges, we proposed a regime-aware and interpretable remote-sensing framework that accounts for heterogeneity in the ground-shaking conditions associated with earthquake-induced secondary hazards. Instead of applying interpretation methods to the entire sample space, we use a PGA-guided multivariate Gaussian mixture model (GMM) to stratify observed hazard samples in a weighted multivariate feature space, with PGA serving as the main physical anchor. The resulting mixture components are treated as PGA-associated statistical regimes. Within each posterior-dominant regime, separate XGBoost models are trained and interpreted using SHAP, enabling comparison of feature contributions under different shaking backgrounds. This design transforms SHAP from a global explanation tool into a regime-wise diagnostic framework for examining how seismic forcing, site conditions, geomorphology, and hydrological constraints vary across hazard types.

Data and methods

SAR coherence and amplitude

SAR coherence has also been used to characterize surface and hydrological conditions relevant to landslide processes (Song et al. 2025b). SAR coherence represents the quality of interferometric measurements by assessing the similarity between two repeated SAR signals, denoted as s_1 and s_2. High coherence usually indicates stable ground conditions. A low coherence is referred to as decorrelation, suggesting ground disturbance, likely caused by ground deformation, vegetation changes, or loss of

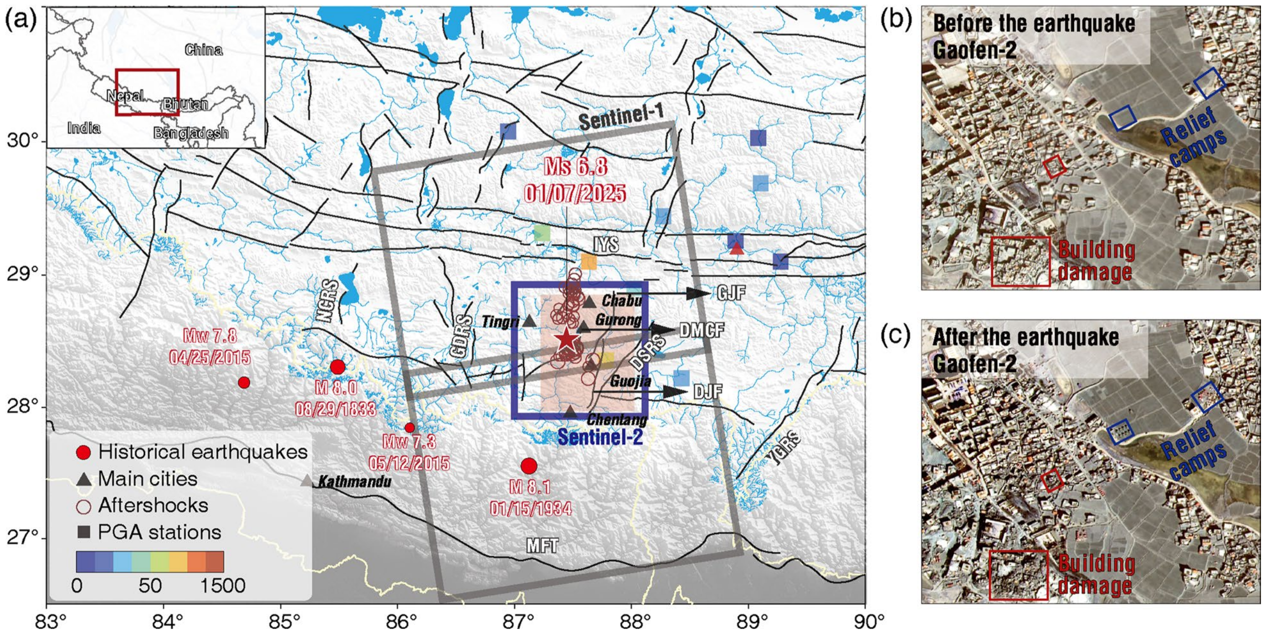

Fig. 1 Tectonic background and earthquake-induced hazards around the epicenter of the 2025 Tibet earthquake. **a** Distribution of the mainshock, aftershocks, and Peak Ground Acceleration (PGA) stations. The gray box shows the footprint of the Sentinel-1 SAR imagery. The blue box shows the footprint of the Sentinel-2 imagery. **b** and **c** are Gaofen-2 images of a damage zone before and after the earthquake, respectively

Table 1 Data sources of the input features for landslides and liquefaction susceptibility models

Type	Variable	Data source	Resolution
Topography	Elevation	SRTM	30 m
	Aspect		
	Slope		
	Roughness		
Earthquake	PGA	Institute of Engineering Mechanics	Site observation
	Vs30	Zhou et al. 2025	30 arcsec
Optical indices	NDWI difference	Sentinel-2	10 m
	NDBI difference		
	NDSI difference		
SAR indicator	GD	Sentinel-1	40 m
	Amplitude	(pre-processed by HyP3)	
Categorical variables	Fault distance	China Earthquake Disaster Prevention Center	GIS output
	River distance	OpenStreetMap	GIS output
	Geological unit type	USGS	GIS shapefile

structural integrity. A precise definition of SAR coherence is given by:

$$\gamma = \frac{|E(s_1 s_2{}^*)|}{\sqrt{E\left(|s_1|^2\right)}\sqrt{E\left(|s_2|^2\right)}} \tag{1}$$

where E denotes the ensemble mean.

The coherence of each pixel in SAR is determined by the expected value derived from multiple SAR images captured at the same time, and this method is not feasible for satellite missions. The ensemble mean can be substituted with a spatial average calculated over a given window. In this case, a moving window is employed to compute the spatially averaged coherence. Consequently, the maximum likelihood coherence is expressed as follows (Seymour & Cumming 1994):

$$\gamma_{\mathrm{MLC}} = \frac{\left|\sum_{m=1}^{M}\sum_{n=1}^{N} s_1(m,n)\, s_2^*(m,n)\right|}{\sqrt{\sum_{m=1}^{M}\sum_{n=1}^{N}|s_1(m,n)|^2 \sum_{m=1}^{M}\sum_{n=1}^{N}|s_2(m,n)|^2}} \tag{2}$$

where M and N are the dimensions of the moving window, and m and n are the two-dimensional image coordinates, respectively.

The European Space Agency (ESA)'s Copernicus Sentinel-1 satellite constellation provides freely available C-band SAR images (Table 1). This study analyzes and applies two frames and six scenes of images related to the ascending Sentinel-1 orbital path, completely covering the epicenter and the primary damage area (Fig. 1). The time interval of images is 12 days. Three SAR images are processed in each frame, namely two before the earthquake and one after (Fig. 1).

In the context of seismic hazards, changes in coherence have proven to be an effective indicator of ground disturbance, which can be a result of landslides, liquefaction, or infrastructure damage. We utilized the HyP3 software to process SAR imagery to compute the coherence, implementing multi-looking factors of 4×1 in the range and azimuth directions. Taking one of the pre-seismic images as the reference, we computed coherence between it and the other two images to obtain pre-event coherence and co-event coherence, respectively. The difference between these two coherence values was then calculated to derive the differential coherence. The coherence-change map is referred to as the ground disturbance map (Song et al. 2025a), a.k.a., damage proxy map (Yun et al. 2015). Variations in spatial baselines and weather conditions can lead to systematic biases in coherence measurements. To mitigate these biases, we further employed histogram matching (Parzen 1962) to reduce biases and removed discontinuities at the edges of the frame, thereby enhancing the quality of the ground disturbance (GD) map. In this study, the GD is used alongside differential SAR amplitude spanning the event to improve the reliability of damage detection and to distinguish between different types of secondary hazards (Fig. 2).

Optical images and derived features

We extracted the Normalized Built-up Index (NDBI), Normalized Water Body Index (NDWI), and Normalized Snow Index (NDSI) using Sentinel-2 multispectral optical images (with a revisit period of 5 days and a spatial resolution of 10 m). The NDBI is calculated using the Shortwave Infrared (SWIR) and Near Infrared (NIR) bands: $\mathrm{NDBI} = \frac{\mathrm{SWIR}-\mathrm{NIR}}{\mathrm{SWIR}+\mathrm{NIR}}$. Higher NDBI values typically indicate a greater likelihood of built-up or other impervious areas (Zha et al. 2003; Zheng et al. 2021). The NDWI is

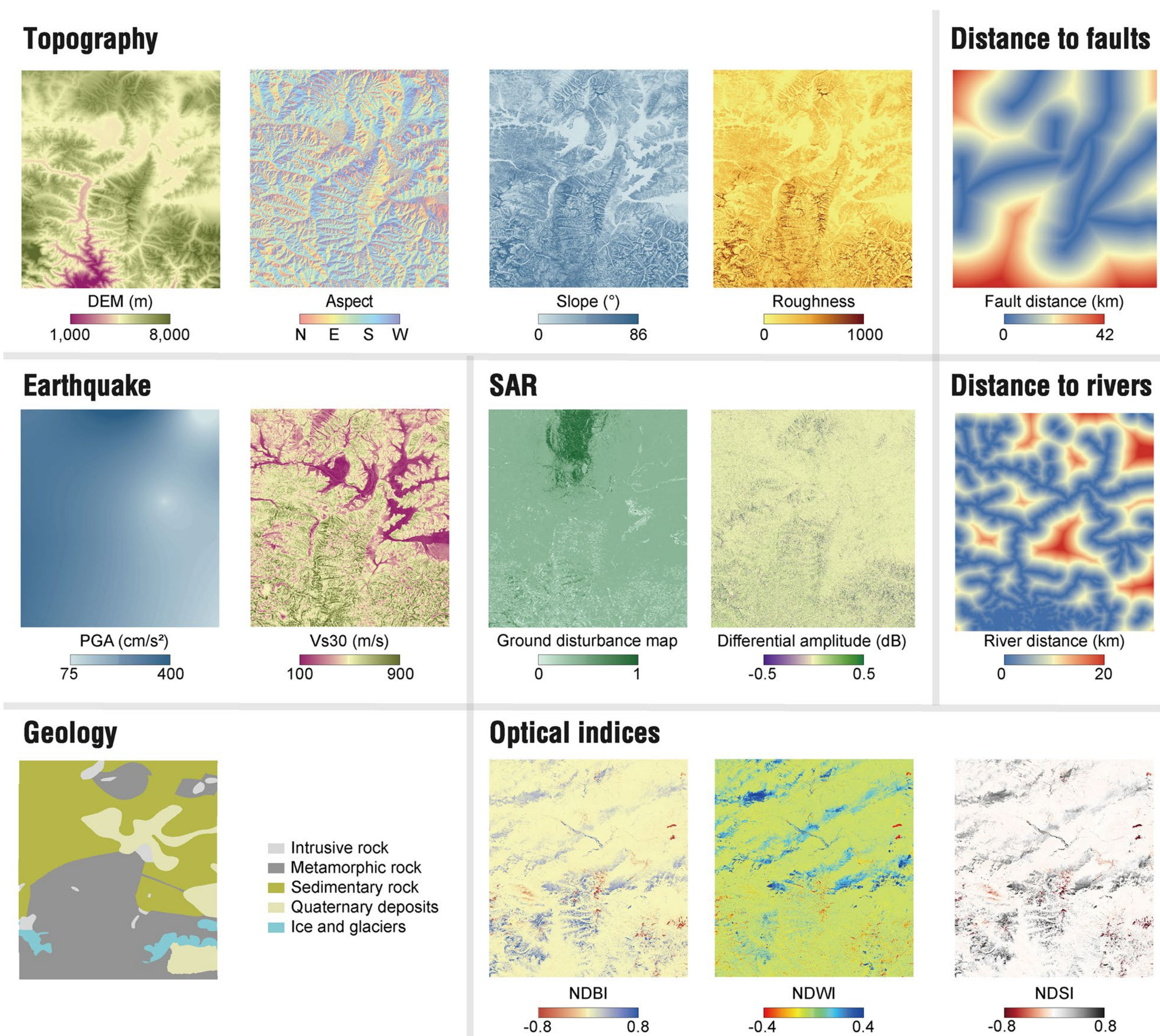

Fig. 2 Multi-source input features for landslides and liquefaction susceptibility modeling

given by $\mathrm{NDWI} = \frac{\mathrm{Green}-\mathrm{NIR}}{\mathrm{Green}+\mathrm{NIR}}$. Higher NDWI values indicate a greater likelihood of water presence (McFeeters 1996). The NDSI is given by $\mathrm{NDSI} = \frac{\mathrm{Green}-\mathrm{SWIR}}{\mathrm{Green}+\mathrm{SWIR}}$. Higher NDSI values indicate the presence of snow or ice, as snow strongly reflects in the visible green wavelengths while absorbing the SWIR (Salomonson & Appel 2004). Then we computed the difference of the indices collected before and after the earthquake to infer GD.

Earthquake properties and geo-environmental features

To conduct a more comprehensive analysis of the possible influencing factors of secondary hazards, we considered various geological environments and seismic characteristics (Fig. 2). The 30-m-resolution digital elevation model (DEM) from the Shuttle Radar Topography Mission (SRTM) represents a collaborative global topographic survey undertaken by NASA and the National Geospatial-Intelligence Agency, and its data has emerged as one of the most extensively utilized global DEMs. The occurrence and distribution of landslides are intricately linked to terrain characteristics such as slope and aspect.

Peak Ground Acceleration (PGA) is a crucial metric in earthquake engineering that quantifies the highest ground acceleration resulting from seismic waves, characterizing the ground shaking intensity. When an earthquake occurs, seismic waves travel through the earth, leading to ground acceleration. PGA indicates the peak instantaneous acceleration experienced by the ground during the seismic event. This measurement is frequently utilized to evaluate the potential effects of earthquakes on engineering structures, and it plays an important role in seismic design and disaster evaluation.

Vs30 refers to the average shear-wave velocity of the soil layer within the top 30 m below the surface, which is a crucial metric for assessing soil layer stiffness and the amplification of seismic ground motions resulting from near-surface seismic activities. Regions characterized by low Vs30 are particularly vulnerable to severe shaking and heightened damage. The 30-arc-second resolution (approximately 900 m) Vs30 has been calibrated with borehole data, with estimation errors decreasing near the measurement sites (Zhou et al. 2025).

Landslide and liquefaction susceptibility mapping

Ground truth

Xu et al. (2025a, b) compared satellite images taken before and after the earthquake, and compiled field investigations to delineate the boundaries of co-seismic landslides and identify the locations of liquefaction. The earthquake-induced hazard labels in vector format contain 2,868 landslide polygons and 406,529 liquefaction sites. For the purposes of enhancing model learning in this study, all labels have been rasterized. The ground truths of hazard occurrences have been sorted out into three categories: 0 for no detection, 1 for landslides, and 2 for liquefaction. Note that there might be multiple liquefaction sites in a single pixel due to the format conversion from point to raster. If there is a liquefaction site within the pixel, we consider that the pixel has undergone liquefaction. A total of 980,699 landslide pixels and 290,093 liquefaction pixels have been sorted out.

Data engineering

To capture the spatial distribution of the hazards and ensure a balanced dataset, we defined the study area based on regions with earthquakes and landslide occurrence. Landslides were mainly distributed near the epicenter and the area from south of Guojia township to Chentang township, and the soil liquefaction occurred primarily in the Pengqu River Valley. Therefore, the study area was constrained to the range of 87°–88°E and 27.6°–29°N (Fig. 1). All input features were projected to the World Geodetic System (WGS84) coordinate system to ensure a consistent geographic reference. The input features were resampled to a 10 m × 10 m grid using the bilinear interpolation approach, resulting in a total of 11,029 × 12,216 pixels. To ensure effective model training and minimize bias, we removed outliers and conducted feature normalization.

We conducted the correlation analysis of the input (Fig. 3). Although these features are designed to reflect distinct surface properties, the optical indices NDSI and NDBI exhibit some correlation. Additionally, slope, Vs30, and surface roughness show intercorrelation. Although there is no direct physical connection among these features, they are all related to surface conditions and the propagation of seismic waves. Slope can aid in the identification of terrain types, which are frequently linked to surface roughness and Vs30. For instance, mountainous regions typically possess steep slopes and are likely to exhibit higher Vs30, as they are predominantly composed of consolidated rocks. Conversely, low-lying flat areas usually have lower Vs30 values due to the presence of soft sediments or loose soils.

Due to the imbalanced ratio of positive (1%) to negative (99%) samples within the affected region, we implemented an undersampling strategy, establishing a 1:5 ratio of positive to negative samples in the dataset (Shao et al. 2020).

To reduce over-optimistic model evaluation caused by spatial autocorrelation among neighboring pixels, a spatially constrained data partitioning strategy was implemented. Instead of random sampling, the study area was divided into spatially independent subsets, and samples were split into training, validation, and test sets with a ratio of 7:2:1. This ensures that geographically adjacent pixels are not simultaneously included in both training

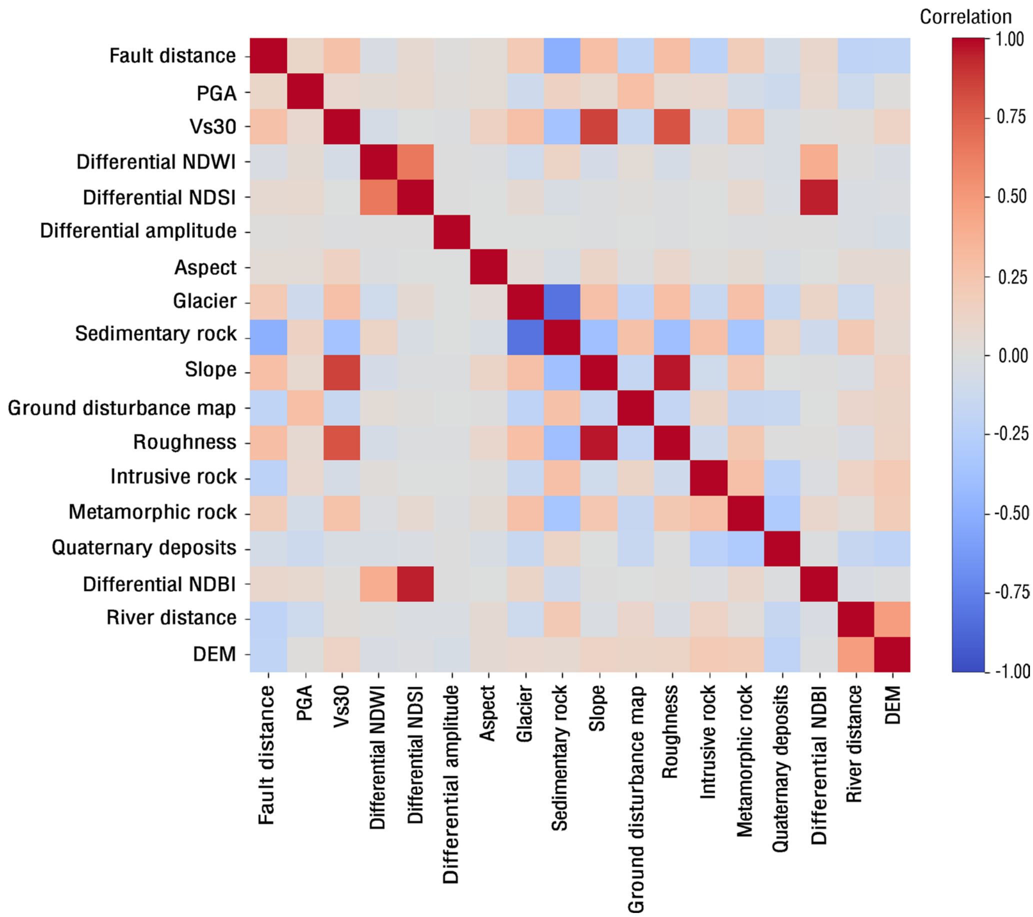

Fig. 3 Correlation heatmap of multisource input features

and testing subsets, thereby providing a more realistic assessment of model generalization under spatial heterogeneity. To reduce the influence of a single spatial split, the partitioning and training process was repeated multiple times, and the results were aggregated to obtain stable performance estimates.

Although the model predictions were generated at a spatial resolution of 10 m, several key predictor variables, such as PGA and Vs30, are available at much coarser spatial resolutions. This scale mismatch is common in regional hazard modeling and does not imply that the model resolves processes at the pixel scale. Instead, the 10 m prediction grid serves as a spatial support for integrating multi-source variables and mapping relative susceptibility. In this context, the model output should be interpreted as the spatial distribution of hazard likelihood conditioned on both coarse-scale seismic forcing and fine-scale environmental variability.

Landslide and liquefaction susceptibility models

With the prepared data samples, We applied three machine-learning methods to assess landslide and liquefaction susceptibility. Specifically, we applied three widely recognized ML techniques: Random Forest (RF), eXtreme Gradient Boosting (XGBoost), and Multilayer Perceptron (MLP).

Random Forest is an ensemble learning algorithm that utilizes a tree-based approach (Breiman 2001; Pal 2005). The algorithm initiates the process by splitting from a root node, which leads to the creation of numerous alternative trees. Users have the flexibility to specify the number of trees, denoted as N, to be produced. Each of the *N* trees in the ensemble casts a vote to determine the most effective classifier for modeling the response variable (He et al. 2021; Oshiro et al. 2012).

XGBoost employs a boosting methodology, comprising several decision trees (Chen & Guestrin 2016). In this model, new trees are incrementally added to address the residuals from prior predictions. After training k trees, the predicted score for a given sample is calculated as the aggregate of the scores assigned by each tree. Each sample is directed to a leaf node within each tree, with each leaf corresponding to a specific score. The overall prediction is derived by summing the scores from all trained trees.

MLP is a variant of Artificial Neural Networks characterized by a fully connected feedforward architecture (Bishop 2006; Rosenblatt 1958; Rumelhart et al. 1986). An MLP is composed of three main components: the input layer, one or more hidden layers, and the output layer. Each neuron within a layer is interconnected with the preceding layer and possesses a set of learnable weights and biases. An MLP contains one or more hidden layers, and the neurons within these layers apply nonlinear transformations using activation functions.

The machine learning models (Random Forest, XGBoost, and Multilayer Perceptron) were trained using the training set. Given the flexibility of these algorithms, hyperparameters were selected based on validation-set performance. The final models were then evaluated on the independent spatial test set using precision, recall, F1-score, Intersection over Union (IoU), pixel accuracy (PA), mean pixel accuracy (mPA), and mean Dice coefficient (mDice).

For each hazard-specific classification task, PA, mPA, and mDice were computed from the confusion matrix. Pixel accuracy (PA) represents the proportion of correctly classified pixels among all evaluated pixels. Mean pixel accuracy (mPA) averages the class-wise accuracies and therefore reduces the dominance of majority classes. The Dice coefficient measures the overlap between predicted and reference pixels for a given class and is calculated as $Dice = \frac{2TP}{2TP+FP+FN}$, where TP, FP, and FN denote true positives, false positives, and false negatives, respectively. The mean Dice coefficient (mDice) is obtained by averaging the Dice coefficient across classes. Together, these metrics evaluate both overall classification correctness and class-wise spatial agreement.

PGA-guided multivariate Gaussian mixture stratification

To account for heterogeneity in the environmental and ground-shaking conditions associated with co-seismic secondary hazards, this study used a PGA-guided multivariate Gaussian mixture model (GMM) to stratify landslide and liquefaction samples before regime-wise interpretation (McLachlan & Peel 2000; Reynolds 2009). PGA directly represents the intensity of seismic shaking and is widely recognized as a primary triggering factor for earthquake-induced landslides and liquefaction. GMM was used as a probabilistic stratification tool to describe major modes in the multivariate feature space. PGA was used as a physical anchor to guide initialization and weighting of the multivariate GMM, so that the resulting components were organized primarily around ground-shaking differences while still reflecting variation in environmental conditions.

For each observed hazard sample, a set of seismic, remote-sensing, geomorphological, hydrological, and site-condition variables was extracted, including PGA, SAR amplitude difference, damage proxy metric, NDBI difference, NDWI difference, surface roughness, DEM, Vs30, distance to fault, and distance to river. These variables were standardized before mixture modeling to reduce the influence of different measurement units. Because PGA represents the primary seismic forcing of earthquake-induced secondary hazards, the standardized PGA variable was assigned a higher weight during GMM fitting. This design allowed the mixture model to remain multivariate while ensuring that the resulting regimes were primarily organized around differences in ground-shaking intensity.

Before fitting the GMM, the PGA distributions of observed landslide and liquefaction samples were examined. The landslide samples showed a bimodal PGA distribution, whereas the liquefaction samples showed a trimodal distribution. These PGA modes were used to define initial low-, intermediate-, and high-shaking classes, which provided physically informed initial component means for the GMM. After initialization, the final mixture parameters were estimated using maximum likelihood in the weighted, standardized multivariate feature space.

Formally, let x_i denote the weighted and standardized feature vector of sample i. The multivariate GMM represents the sample distribution as:

$$p\left(\mathrm{PGA}\right) = \sum_{k=1}^{K} \pi_k N(x_i|\mu_K, \sum_{k}) \qquad (3)$$

where K is the number of mixture components, π_k is the mixture weight of component k, and μ_k and $\sum_k$ are the mean vector and covariance matrix of that component, respectively. Two components were used for landslides and three components for liquefaction, consistent with the observed PGA distributional modes.

For each sample, the GMM estimates posterior membership probabilities for all mixture components. The component with the highest posterior probability was used as an operational regime label for subsequent XGBoost modeling and SHAP interpretation. Samples near overlaps between Gaussian components represent transitional cases with uncertain regime membership. Therefore, the resulting regimes are referred to as PGA-guided multivariate regimes.

The purpose of this stratification is to reduce the averaging effect that may occur when all hazard samples are interpreted together. The physical meaning of each regime is not assumed from the GMM alone, but is evaluated through the subsequent regime-wise SHAP analysis, which compares how seismic forcing, topographic setting, hydrological conditions, site properties, and near-fault effects contribute to model predictions under different ground-shaking backgrounds.

SHAP-based model interpretation

To move beyond predictive performance and examine model-inferred feature contributions, this study employs SHAP to interpret the machine learning models trained within each PGA-derived regime. SHAP is a model-agnostic explanation framework grounded in cooperative game theory (Lundberg & Lee 2017), which decomposes a model's prediction into additive contributions from individual input features.

For a given sample x, the model output f(x) is expressed as:

$$f\left(x\right)=\phi_0+\sum_{j=1}^{M}\phi_j \tag{4}$$

where ϕ_0 represents the expected model output over a background distribution, M is the number of input features, and ϕ_j denotes the SHAP value corresponding to the contribution of the j^{th} feature. Positive SHAP values indicate that a feature increases the model's tendency toward hazard occurrence, whereas negative values indicate a suppressing effect.

In this study, SHAP values were computed using the TreeExplainer algorithm, which provides an efficient and exact solution for XGBoost. SHAP analysis was performed separately for each PGA regime using the corresponding regime -specific XGBoost model, ensuring that feature contributions are interpreted conditional on a consistent seismic intensity regime.

Results

Susceptibility model performance

Landslides are mainly concentrated within and to the south of the Dengme Co Basin, especially in the southern edge of the Guojia Basin. Liquefaction mainly occurs in the Pengqu River Basin and the southwestern edge of the demarcation basin, showing certain topographic and hydrological control characteristics (Fig. 4a). This distribution pattern reflects the differences in response to geological environments.

The results generated by the RF and XGBoost models are highly consistent with the ground truth, suggesting that the two integrated learning models have strong capabilities in extracting key geological and topographic factors in landslides and liquefaction (Fig. 4b, c). Although the MLP can still effectively identify high susceptibility areas in terms of the overall spatial trend, its sensitivity to landslides and liquefaction is slightly overestimated in some marginal areas (Fig. 4d). Nevertheless, the MLP model still performs well in the prediction of high-risk areas for landslides and liquefaction.

The post-earthquake hazard products from USGS provide global-scale predictions for co-seismic landslides and liquefaction based on empirical models. Our comparison shows a good spatial consistency in liquefaction-prone areas between USGS outputs and our susceptibility model, particularly along the Pengqu River and low-lying basins. On the other hand, USGS-predicted landslide susceptibility extends more broadly, overestimating hazard in distant regions. This discrepancy likely reflects the limitations of global empirical models when applied to tectonically complex terrains in the southern Tibetan Plateau.

To further verify the comprehensive performance of the model, this study also calculated multiple evaluation indicators, including precision, recall, F1-score, IoU, mDice, PA, and mPA (Table 2). These indicators provide a comprehensive assessment of the applicability and robustness of each model in the actual assessment of earthquake-induced landslides.

Contributions of features to landslide and liquefaction

We examined the significance of various features using the permutation importance approach. Specifically, the values of a particular feature are randomly rearranged, and the model generates predictions based on the altered dataset. By comparing the predicted outcomes with the actual target values, the loss function is calculated to measure the increase in prediction error resulting from the random shuffling. The extent of performance decline indicates the importance of the respective feature. By repetitively executing this procedure across all features, their relative significance can be quantified (Fig. 5).

For landslides (Fig. 5a–c), three models show a high degree of consistency in the ranking of feature importance. PGA and DEM are consistently among the top predictors across all models, indicating that co-seismic landslide predictions were primarily associated with ground-motion intensity and topographic variables.

In contrast, liquefaction exhibits a "multi-factor coupling" behavior: its occurrence depends not only on seismic intensity but also on local environment and geomorphological characteristics. The differences among models in key controlling factors for liquefaction further reflect the nonlinear complexity of the liquefaction process in feature space.

Discussion

PGA-guided multivariate mixture regimes

The feature-importance rankings of the liquefaction susceptibility models reveal more dispersed contributions across different algorithms than those of landslides, indicating that liquefaction is jointly associated with

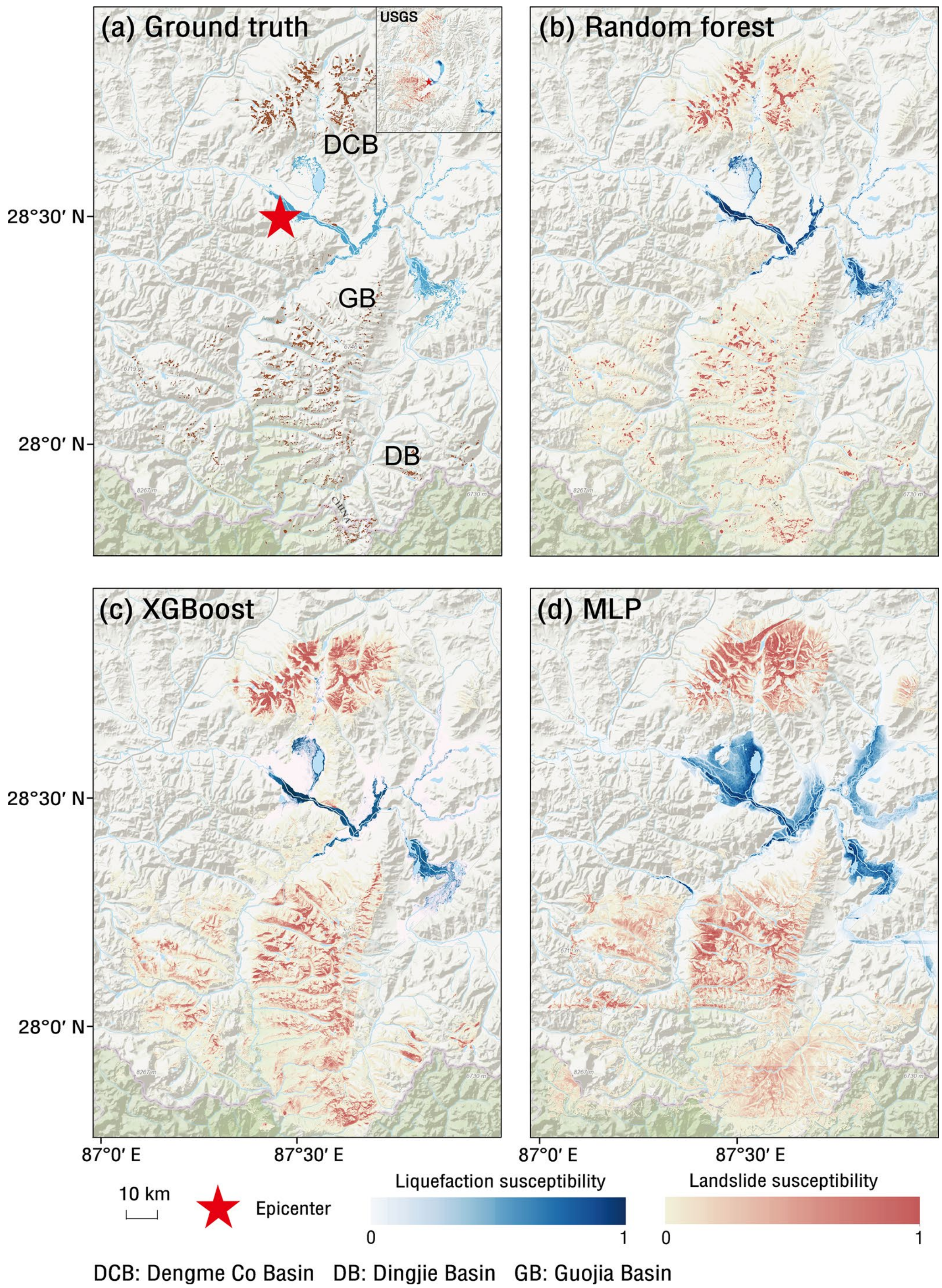

Fig. 4 Landslide and liquefaction susceptibility assessment. **a** Spatial distribution of landslides and liquefaction (The result of USGS is in the upper right corner). **b**–**d** The susceptibility prediction results of the RF, XGBoost, and MLP models

multiple environmental and seismic factors. This contrast suggests that, at the scale of the full sample set, using a single unified interpretation may obscure differences in the relationships between hazard occurrence and controlling variables. Therefore, we further examined the feature distributions of observed landslide and liquefaction samples, with particular attention to PGA because it represents the primary ground-shaking input for coseismic secondary hazards (Fig. 6).

Inspection of the PGA distributions indicates that ground-motion intensity does not follow a single continuous pattern for either hazard type. Instead, the PGA histogram for landslides samples shows a clear bimodal structure, which motivates a two-zone partition (Fig. 6b).

Table 2 Comparison of machine learning models (LS = Landslide, LQ = Liquefaction). Bold values indicate the results of the model selected for subsequent analysis

Metric	XGBoost (LS)	XGBoost (LQ)	MLP (LS)	MLP (LQ)	RF (LS)	RF (LQ)
Precision	**0.9208**	**0.8683**	0.7849	0.7163	0.8668	0.8452
Recall	**0.9518**	**0.9298**	0.6623	0.7951	0.8888	0.9064
F1 score	**0.9312**	**0.8980**	0.7184	0.7537	0.8776	0.8747
IoU	**0.9130**	**0.8149**	0.5605	0.6048	0.7820	0.7774
mDice	**0.9071**	**0.9571**	0.8069	0.8069	0.9091	0.9091
PA	**0.9370**	**0.9870**	0.9135	0.9135	0.9582	0.9582
mPA	**0.9202**	**0.9702**	0.8050	0.8050	0.9221	0.9221

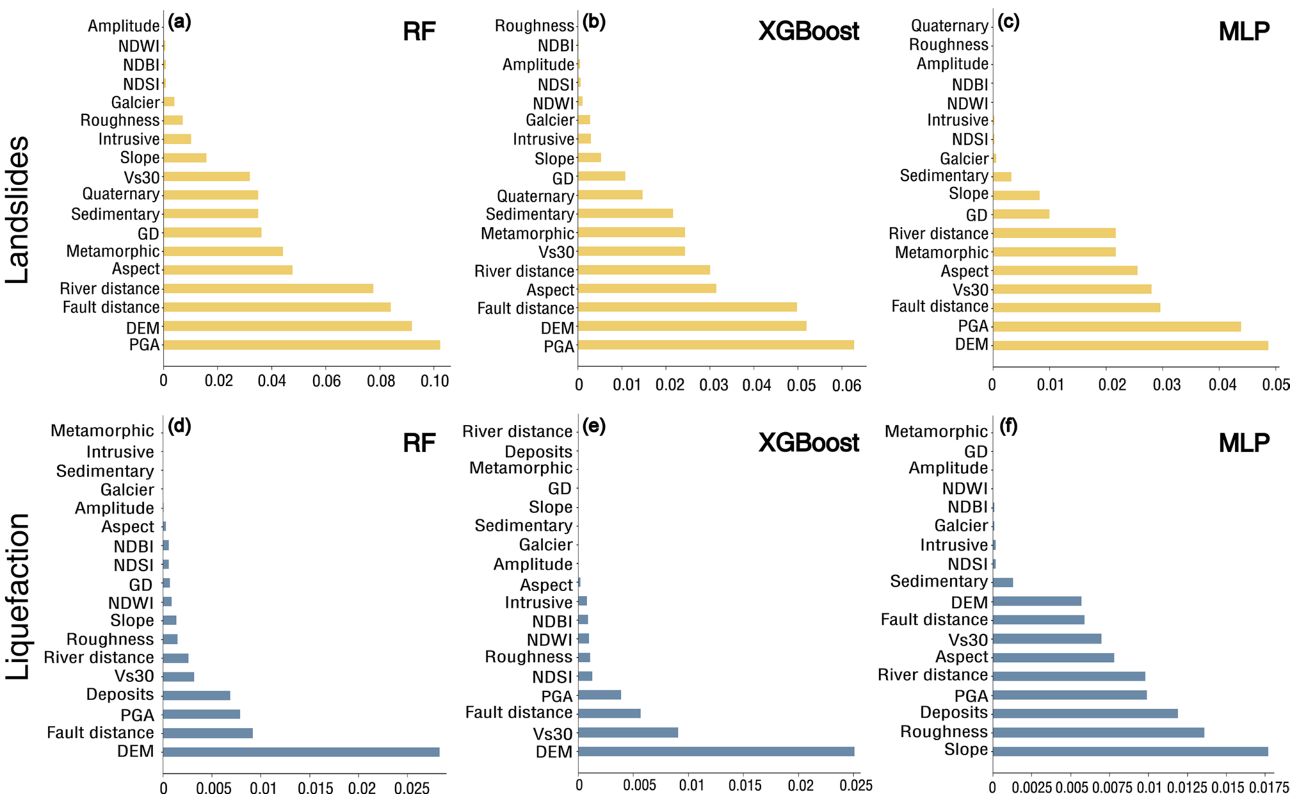

Fig. 5 Rank of feature importance in landslide and liquefaction susceptibility assessment. **a**–**c** feature importance of the landslide in RF, XGBoost, and MLP, respectively. **d**–**f** feature importance of the liquefaction in RF, XGBoost, and MLP, respectively

Consistent with this, the landslide regimes separate into a moderate-shaking zone (Zone I; ~ 240–300 cm/s^2) and a high-shaking zone (Zone II; ~ 300–360 cm/s^2), with limited overlap around the transition. In contrast, liquefaction samples exhibit a pronounced multimodal PGA distribution (Fig. 6c), supporting a multi-zone representation: a low-PGA liquefaction zone (Zone III; ~ 170–220 cm/s^2), an intermediate zone (Zone IV; ~ 230–270 cm/s^2), and a high-PGA zone (Zone V; ~ 280–320 + cm/s^2), where the peaks suggest distinct shaking regimes associated with different liquefaction settings. These regimes should be interpreted as statistical strata associated with different shaking and environmental contexts.

Based on these considerations, we introduced a PGA-guided multivariate GMM to stratify landslide and liquefaction samples (Fig. 7). PGA was used as the main physical anchor through both initialization and weighting, while other variables, including remote-sensing indicators, topographic attributes, site-condition variables, and hydrological or tectonic proximity metrics, were also included in the standardized multivariate feature space. Therefore, the resulting regimes are not equivalent to simple PGA thresholds. Instead, they reflect the joint distribution of ground shaking and environmental conditions, with PGA exerting a stronger organizing influence. GMM estimates a weighted combination of Gaussian components in feature space, and samples near the overlap between components may have uncertain

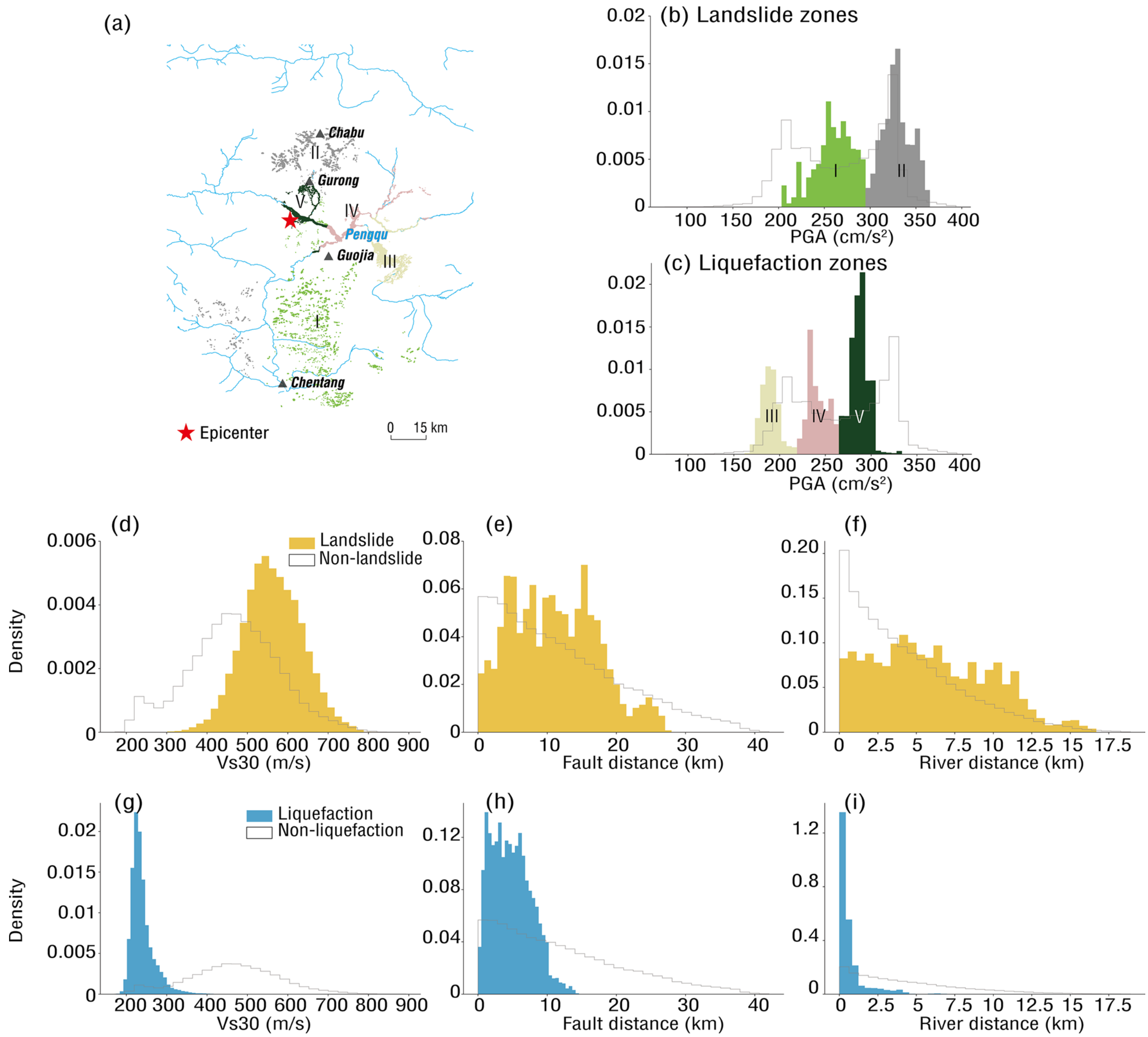

Fig. 6 Distributions of important features compared between landslide/liquefaction and background non-landslide/non-liquefaction samples. **a** The distribution of landslides and liquefaction, with different colors representing different regimes. **b**–**c** The histogram distributions of landslides and liquefaction on the PGA, and the colors correspond to the distributions in (**a**). **d**–**f** respectively represent the histogram distributions of the landslide in RF, XGBoost, and MLP. **g**–**i** respectively represent the histogram distributions of the liquefaction in RF, XGBoost, and MLP

membership. In this study, the maximum-posterior component was used as an operational regime label for subsequent XGBoost and SHAP analyses.

Regime-wise SHAP-based attribution interpretation

To examine how model-inferred controlling factors vary across different shaking–environmental backgrounds, we trained XGBoost models separately within each posterior-dominant PGA-guided regime and interpreted the model outputs using SHAP. SHAP decomposes model predictions into additive feature contributions, enabling a quantitative comparison of the direction and magnitude of each variable's influence on predicted hazard occurrence. This regime-wise interpretation reduces the confounding effects of sample heterogeneity in overall explanations and allows the controlling mechanisms under different PGA backgrounds to be identified more clearly. To enhance the model robustness and eliminate interfering factors, we conducted a screening of the features based on the previous correlation analysis and feature importance. We retained ten features that were important as the final input.

Regime-dependent attribution patterns for liquefaction

The SHAP results for liquefaction indicate clear differences in the relative contributions of controlling variables

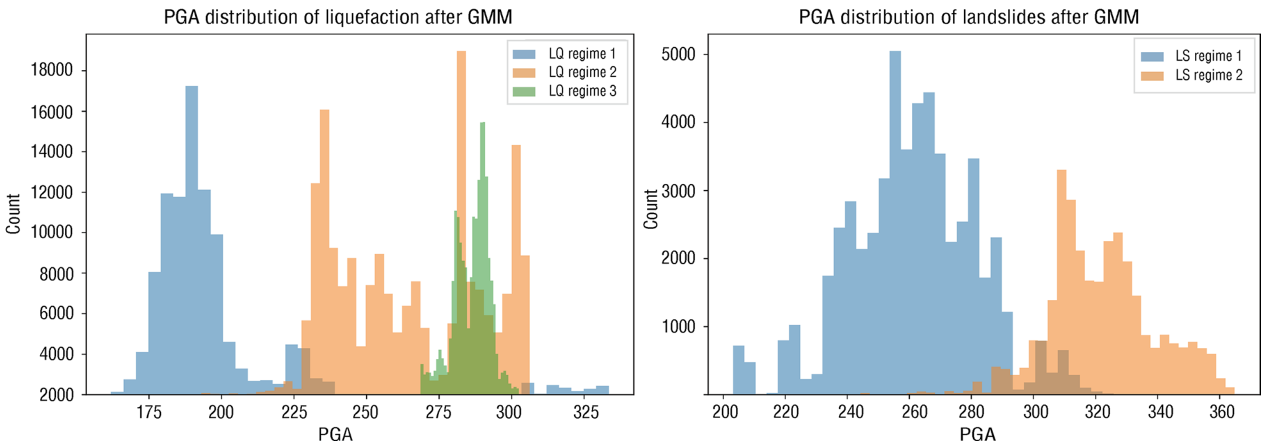

Fig. 7 Histogram distribution after PGA-guided GMM

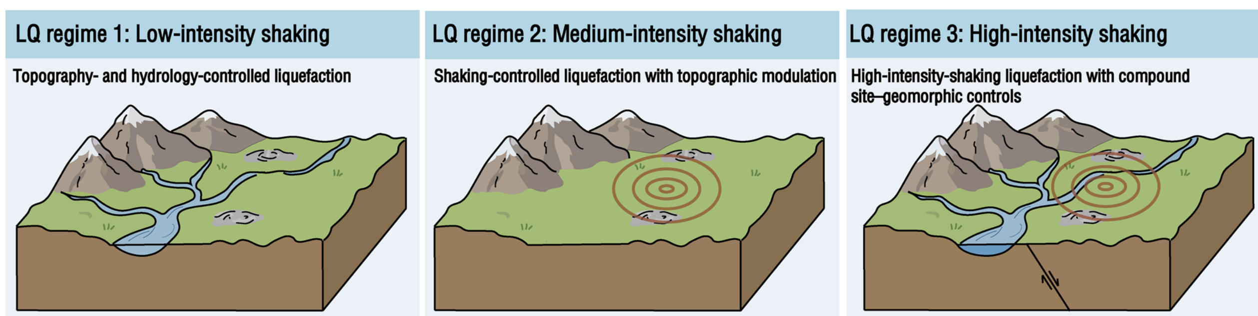

Fig. 8 Control factors for liquefaction under different vibration intensities

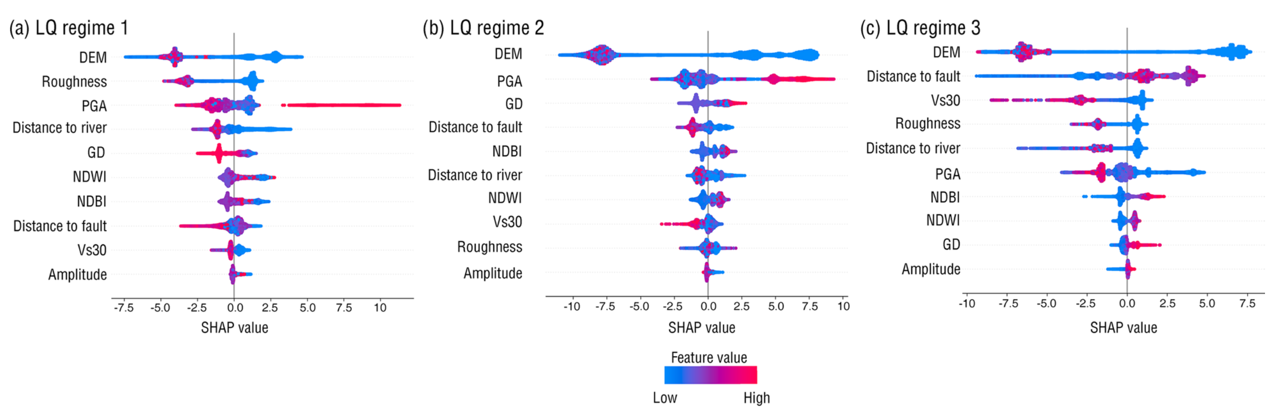

Fig. 9 SHAP of liquefaction in different regimes

across the three PGA-guided regimes, suggesting that the model-inferred conditions associated with liquefaction vary under different shaking backgrounds (Fig. 8).

(1) Topography- and hydrology-controlled liquefaction

As shown in Fig. 9a, under the low PGA regime (LQ Regime 1), DEM and roughness exhibit the widest SHAP value ranges, suggesting that the predicted liquefaction occurrence in this regime is strongly associated with topographic and geomorphic setting. Lower elevations and lower surface roughness (i.e., flatter and more homogeneous surfaces) tend to increase the predicted liquefaction probability, consistent with geomorphic units such as river valleys and plains where fine-grained, loose sediments concentrate, and water availability is generally high. PGA still contributes positively, but its relative contribution is weaker than that of topographic and surface-condition variables. Distance to river and NDWI also show moderate contributions, indicating that hydrological conditions may modulate liquefaction susceptibility under relatively lower shaking levels (Fig. 10).

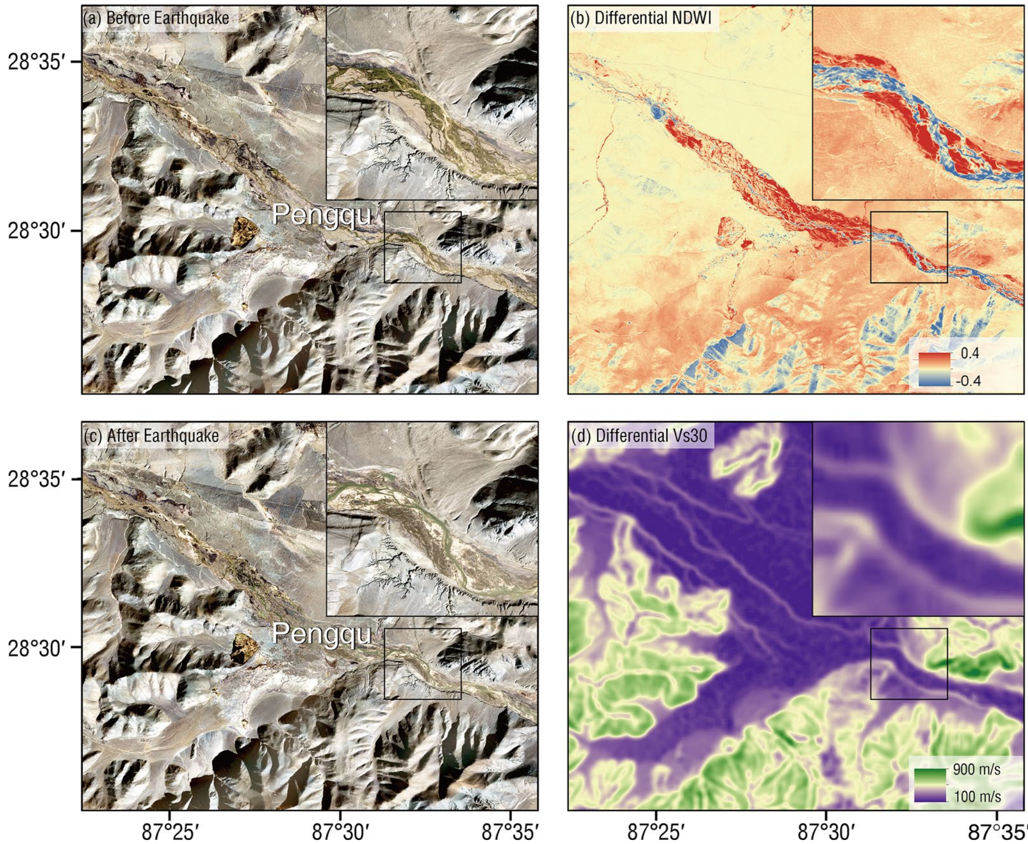

Fig. 10 Liquefaction near the Pengqu River

(2) Shaking-controlled liquefaction with topographic modulation

Figure 9b shows that under a medium PGA regime (LQ Regime 2), the contribution of PGA increases markedly and exhibits a strongly asymmetric SHAP distribution: higher PGA values exert a more pronounced positive push on liquefaction occurrence probability, suggesting that this regime approaches a more shaking-dominated triggering regime. Meanwhile, DEM remains highly important, suggesting that even under stronger shaking, the spatial manifestation of liquefaction is still strongly modulated by topographic position (valleys/plains versus higher terrain). The contributions of distance to fault and the damage proxy metric also increase in this regime, implying a tighter spatial coupling between near-fault tectonic effects, co-seismic ground disturbance, and liquefaction occurrence. Compared with Regime 1, the effects of hydrological and geomorphological indices persist but become relatively weaker, reflecting the elevated weight of triggering conditions under strong ground motion.

(3) High-intensity-shaking liquefaction with compound site–geomorphic controls

Under the high-PGA regime (LQ Regime 3), multiple variables—including DEM, distance to fault, Vs30, roughness, and distance to river—exhibit comparable SHAP value ranges, with no single variable showing an overwhelmingly dominant contribution (Fig. 9c). In this setting, ground-motion intensity is closer to a broadly satisfied triggering background in space, and the spatial expression of liquefaction is more likely to be amplified through several physically plausible alternative pathways. Specifically, site softness (low Vs30), near-fault effects, geomorphic–depositional conditions (low DEM and low roughness), and hydrological proximity to rivers can each substantially increase liquefaction likelihood for different subsets of samples, resulting in a more balanced multivariable importance pattern in SHAP. This pattern suggests that the inferred mechanism is not controlled by a single dominant predictor, but by several competing or complementary controls whose relative influence may vary across samples.

Regime-dependent attribution patterns for earthquake-induced landslides

The SHAP explanation of earthquake-induced landslides shows that the differences in landslide mechanisms corresponding to the two types of PGA regimes are more

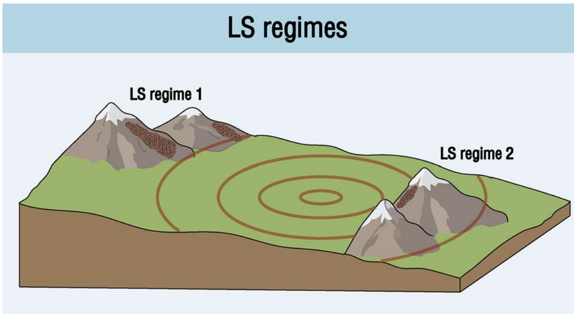

Fig. 11 Control factors for landslides under different vibration intensities

clearly manifested, reflecting the dominant roles of the shaking and topography (Fig. 11).

(1) Shaking-induced landslides in geomorphologically sensitive terrain

As shown in Fig. 12a, within the lower-PGA regime, PGA exhibits the largest SHAP magnitude and is dominated by positive contributions, suggesting that landslides in this regime are more strongly influenced by ground-motion intensity and may be interpreted as a relatively shaking-dominated landslide type. Roughness and DEM also contribute substantially, suggesting that highly dissected terrain (high roughness) and specific topographic settings increase slope instability susceptibility. Distance to fault shows a strong influence in this regime as well, implying that near-fault strong shaking, rupture-related effects, or localized ground disturbance play an important role in triggering landslides. Overall, this regime is most consistent with a combined triggering mode characterized by strong shaking and terrain predisposition.

(2) Landslide predictions associated with topographic and structural setting

In the high-PGA regime, DEM becomes the most important variable (Fig. 12b), and elevation shows a stronger association with landslide susceptibility, indicating that the spatial distribution of this regime is more strongly associated with topography and long-term landscape evolution. The SHAP distribution of PGA is comparatively more symmetric, suggesting that seismic shaking in this regime may have already exceeded a basic triggering threshold; subsequently, whether landslides occur depends more on the "preconditioned susceptibility" imposed by terrain structure and material properties. Meanwhile, roughness, distance to fault, and the damage proxy metric jointly contribute, highlighting the roles of structural control and potentially unstable discontinuities or loose materials. Under the high-relief and tectonically active setting of the southern Tibetan Plateau, such a pattern is consistent with slopes that are already close to failure and are then differentially activated by strong shaking. Overall, this regime is more consistent with a mechanism in which predisposing conditions dominate, in line with observations of landslide displacement behavior in active fault zones (Shi & Hu 2023).

Comparative interpretation and implications

Overall, the regime-wise SHAP analysis suggests that both liquefaction and landslide predictions exhibit strong heterogeneity in their model-inferred attribution patterns within a single seismic event. The PGA-guided multivariate GMM provides a statistical stratification of hazard samples in a weighted multivariate feature space, with PGA serving as the main physical anchor. The

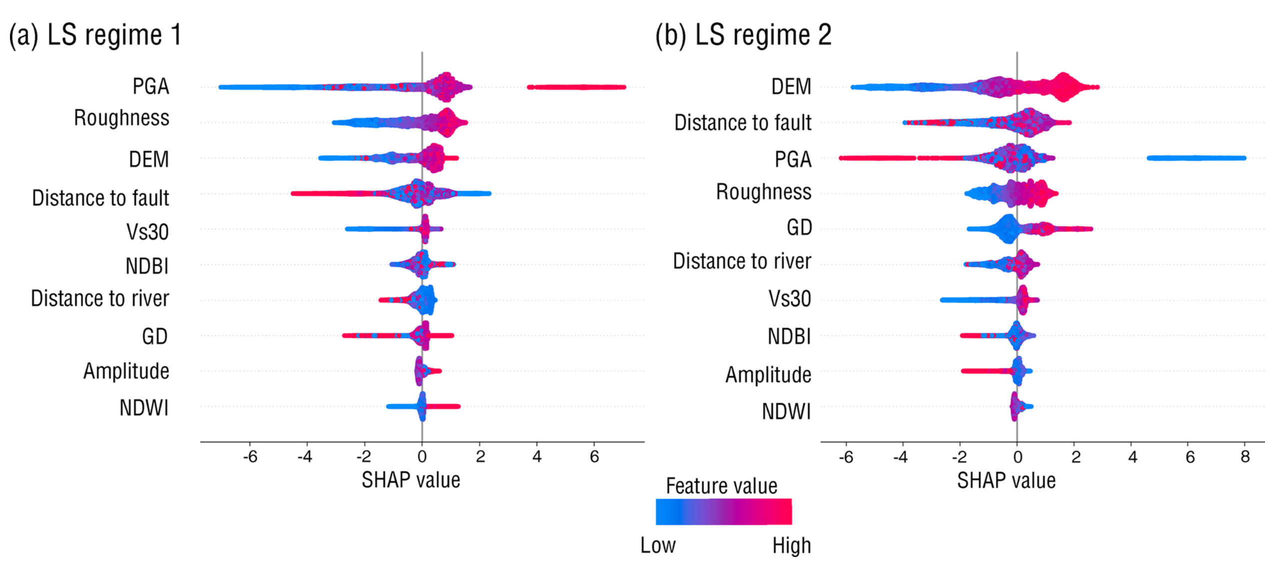

Fig. 12 SHAP of landslide in different regimes

geoscientific interpretation arises from comparing SHAP attribution patterns across these regimes and examining whether they are consistent with known seismic, geomorphic, hydrological, and site-condition processes.

The results suggest that landslides show a relatively consistent coupling between ground shaking and terrain predisposition. PGA, DEM, roughness, and distance to fault remain important across regimes, indicating that landslide occurrence in the Dingri earthquake was closely associated with both seismic forcing and geomorphologically susceptible terrain. In contrast, liquefaction exhibits a more distributed and regime-dependent attribution structure. Under lower shaking conditions, topographic and hydrological variables are more prominent, whereas under stronger shaking conditions, site conditions, near-fault effects, and depositional or fluvial settings jointly influence model predictions. This contrast is consistent with the different physical requirements of the two hazard types: landslides require slope instability in steep terrain, whereas liquefaction depends on the combination of shaking, loose or saturated materials, and local depositional or hydrological environments. GMM regimes summarize dominant modes in the weighted feature space and may overlap, while SHAP values quantify feature attributions within trained XGBoost models. The consistency between the regime-wise attribution patterns and established geoscientific understanding suggests that the proposed framework provides a useful diagnostic approach for comparing earthquake-induced secondary hazards under spatially heterogeneous conditions.

From a disaster-management perspective, the regime-wise interpretation provides information beyond susceptibility mapping alone. Areas where landslide predictions are dominated by strong shaking and steep terrain may require rapid slope-failure screening, whereas liquefaction-prone river valleys with low Vs30 and high water-related indicators may require focused inspection of roads, bridges, settlements, and lifeline infrastructure. Therefore, the proposed framework can support post-earthquake prioritization of field investigation and risk mitigation in remote high-altitude environments.

Conclusions

This study presents a regime-aware and interpretable framework for analyzing earthquake-induced landslides and liquefaction using multi-source remote sensing data, demonstrated with the 2025 Dingri earthquake in the southern Tibetan Plateau. By integrating PGA-guided multivariate stratification with regime-specific machine learning and SHAP interpretation, the approach enables comparison of model-inferred attribution patterns under different shaking and environmental backgrounds. The results suggest that landslide and liquefaction predictions are associated with different combinations of seismic forcing and environmental conditions, rather than being adequately represented by a single uniform attribution pattern. The proposed framework improves the geoscientific interpretability of data-driven hazard models by linking model predictions to regime-dependent feature associations. Overall, the study provides evidence that accounting for spatial heterogeneity can improve the interpretation of earthquake-induced secondary hazards within a single event. While this study focuses on the Dingri earthquake, the framework offers a structured basis for future evaluation in other seismic settings.

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Author contribution

L. F. and Y. X. wrote the manuscript text, processed the data, and conducted the experiments. L. F., Y. X., and H. X. prepared the figures. H. X. provided the ideas, was responsible for the project funds, and managed the project. X. Y. provided data and conducted the feasibility analysis. A. also conducted the feasibility analysis. All the authors reviewed the manuscript.

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Data availability

The datasets supporting the conclusions of this article are available in the Zenodo repository, https://doi.org/10.5281/zenodo.20108768. The original landslide and liquefaction inventory used to generate the ground-truth labels is available from Xu et al. (2025).

Declarations

Competing interests

The authors declare there are no conflicts of interest for this manuscript.

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